Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/15/2015
PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/28/2012	**End Date:**	06/28/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kohel, James (Jet Propulsion Laboratory) Yu, Nan (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL 1458850		
Performance Goal No.:			
Performance Goal Text:			
	The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's (European Space Agency) quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter.		

Task Description:	The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at UC Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand two things: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein's principle of equivalence? We have made progress on answering these questions that have been pursued both theoretically and experimentally.
Task Progress:	We investigated theoretical and experimental aspects of planned missions for quantum tests of the equivalence principle suing atom interferometry in space. Deviations from the equivalence principle may arise from ultra-light particles that might make up all or part of dark matter and dark energy. If dark energy — which drives the accelerated expansion of the universe — consists of a light scalar field, it might be detectable as a "fifth force" between normal-matter objects, in potential conflict with precision tests of gravity. Chameleon fields and other theories with screening mechanisms, however, can evade these tests by suppressing the forces in regions of high density, such as the laboratory. We have shown theoretically and experimentally how atom interferometers can search for these fields. This is very interesting, because other versions of such fields might require space missions to be detected, being intrinsically sup-pressed anywhere near the Earth. Other experimental work was concerned with reducing systematic errors of atom interferometers. Our experiment is one of the few worldwide with sufficient sensitivity to probe these effects on the scales relevant for future missions. This allowed us to develop theoretical models and verify them by direct comparison with the experiment, and point out ways to mitigate these systematic effects. We improved interferometer contrast to about 30% at a pulse separation of 150 ms by improving velocity selection and reviving the Coriolis force compensation. New interferometer geometries that use Bloch oscillations to enhance the signal and suppress systematic effects were also demonstrated. As a result, we were able to point out interferometer geometries which are insensitive to the systematic effects studied. Alternatively, we reduced the effects ~1000 times by introducing feedback methods. As part of this project, we initiated a detailed study of a future state-of-the-art interferometer in space. The work resulted in one publication in Physical Review Letters and two submitted manuscripts.
Bibliography Type:	Description: (Last Updated: 01/29/2019)
Articles in Other Journals or Periodicals	Hamilton P, Jaffe M, Haslinger P, Simmons Q, Müller (Mueller) H, Khoury J. "Atom-interferometry constraints on dark energy." Submitted for publication, February 2015. Cornell University Library arXiv:1502.03888v2 [physics.atom-ph] ; http://arxiv.org/abs/1502.03888v2 , Feb-2015
Articles in Other Journals or Periodicals	Estey B, Yu C, Müller (Mueller) H, Kuan P-C, Lan S-Y. "High sensitivity, low-systematics atom interferometers using Bragg diffraction and Bloch oscillations." In revision, June 2015. Cornell University Library arXiv:1410.8486v3 [physics.atom-ph] ; http://arxiv.org/abs/1410.8486 , Jun-2015
Articles in Peer-reviewed Journals	Hamilton P, Jaffe M, Brown JM, Maisenbacher L, Estey B, Müller (Mueller) H. "Atom interferometry in an optical cavity." Physical Review Letters. 2015 Mar 11;114(10-13):100405. http://dx.doi.org/10.1103/PhysRevLett.114.100405 , Mar-2015